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Quiz 6

1 Core Pseudocode of the actor-critic algorithm:

1. Initialize

1. Initialize policy network (Actor) $\pi_\theta(a \mid s)$ with parameter θ
2. Initialize value network (Critic) $V_\phi(s)$ with parameter ϕ
3. Set learning rates $\alpha_\theta, \alpha_\phi$, discount factor γ

2. For each episode:

Reset environment to initial state s
while not terminal state **do**
    Actor selects action $a \sim \pi_\theta(\cdot \mid s)$
    Execute a, observe reward r, next state s'
    Compute TD error (critic error): $\delta = r + \gamma V_\phi(s') - V_\phi(s)$
    Update Critic (minimize MSE of TD error): $\phi \leftarrow \phi + \alpha_\phi \cdot \delta \cdot \nabla_\phi V_\phi(s)$
    Update Actor (maximize expected reward via policy gradient): $\theta \leftarrow \theta + \alpha_\theta \cdot \delta \cdot \nabla_\theta \log \pi_\theta(a \mid s)$
    $s \leftarrow s'$
end while

3. Repeat until convergence: